
MS1411: MATHEMATICAL STATISTICS (WEEK 2)

Bayes' Theorem; Discrete Random Variables; Mean, Variance, and Standard Deviation; Discrete Distributions

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Course: Based on Walpole et al., "Probability & Statistics for Engineers & Scientists," 9th edition (Chapters 2.7, 3.1–3.2, 4.1–4.4, 5.1–5.5)



MOTIVATION AND REAL-WORLD CONTEXT

- **Why These Topics Matter**

- Bayes' Theorem underlies many modern machine-learning applications (e.g., spam filtering, medical diagnostics).
- Discrete random variables model counts or finite outcomes: defect counts, heads/tails, arrivals of events, etc.
- Mean and variance characterize the “location” and “spread” of these random phenomena.
- Binomial, Poisson, and other discrete distributions arise constantly in engineering (quality control, reliability) and science (genetics, epidemiology).

BAYES' THEOREM: INTUITION AND FORMULA

- **Reference:** Walpole et al., 9th ed. (Chapter 2.7)
 - **Why We Need Conditional Probability**
 - Often we want to know $P(A|B)$, the probability of some event A (e.g., “disease present”) given that another event B has occurred (e.g., “test result positive”).
 - Conditional Probability Recall: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
 - **Bayes' Theorem** (The “Inversion” Formula)
 - $$P(A|B) = \frac{P(B | A) * P(A)}{P(B | A) * P(A) + P(B | A^c) * P(A^c)}$$
 - Reads: Probability of A given B equals the probability of B given A times the “prior” probability of A , divided by all ways B can happen (whether A happens or not).
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BAYES' THEOREM: INTUITION AND FORMULA

- **Interpretation**

- Prior vs. Posterior: We start with a “prior” belief $P(A)$. We observe new evidence B . Bayes' Theorem updates this to a “posterior” probability $P(A|B)$.
- Why It's Powerful: It lets us re-evaluate how likely an event is when we learn something new.

- **Common Applications**

- **Medical Testing:** Probability a patient truly has a disease given a positive test (false positives/false negatives matter!).
 - **Machine Learning:** Classifiers update their predictions when new data arrive.
 - **Quality Control:** Probability an item is defective if an inspection “flags” it.
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BAYES' THEOREM EXAMPLE: DEFECT DETECTION

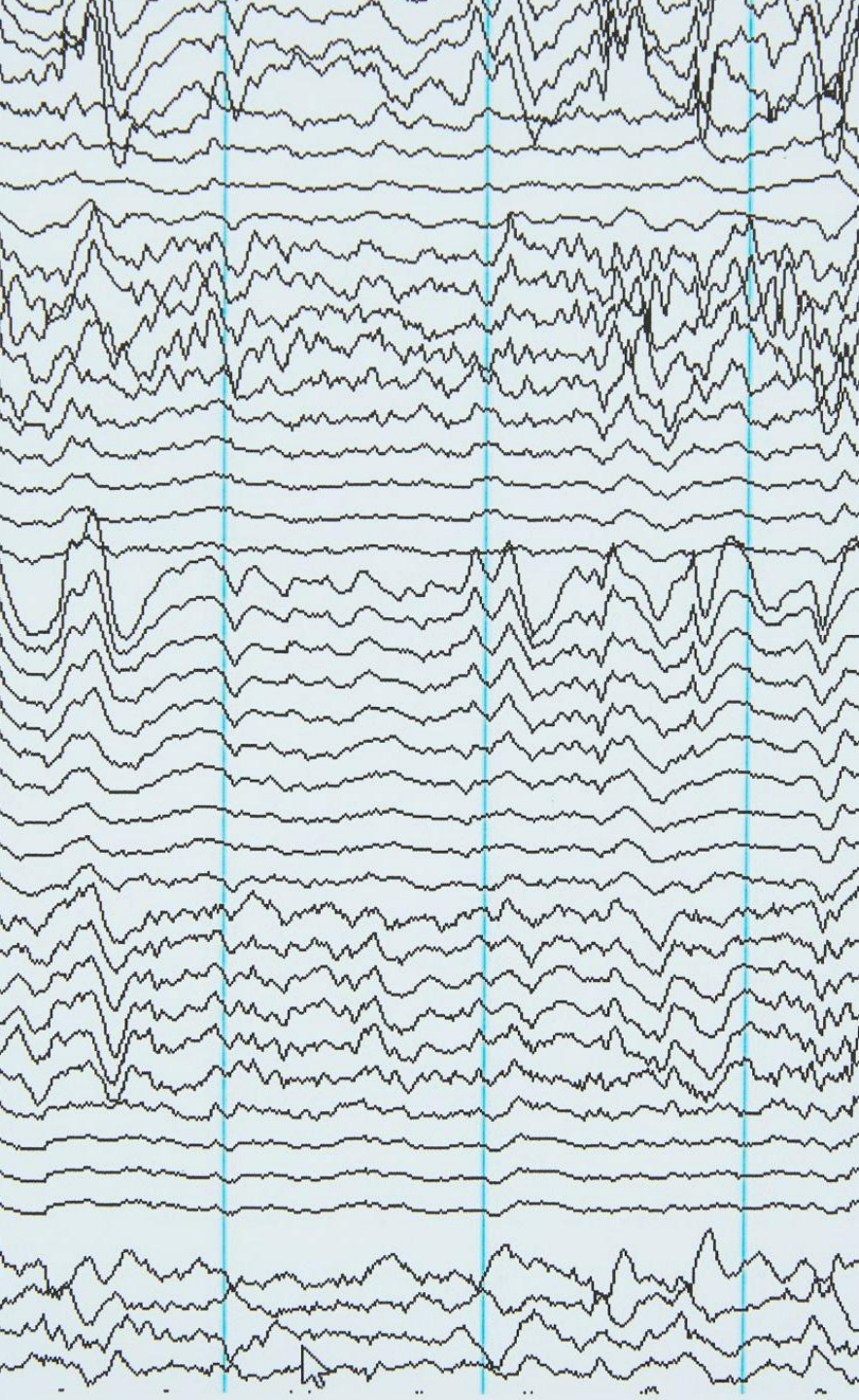
- **Manufacturing Quality-Control Scenario**

- Let event A = “item is truly defective.”
- Let event B = “inspection test indicates defective.”

- **Given Info**

- $P(A)=0.01$: Only 1% of items are actually defective (this is the “prior”).
- $P(B|A)=0.90$: If the item is defective, the test is 90% likely to flag it (true positive rate).
- $P(B|A')=0.05$: If the item is not defective, there is a 5% chance the test incorrectly flags it (false positive).

- Using Bayes' Theorem:
$$P(A | B) = \frac{(0.9)(0.01)}{(0.9)(0.01)+(0.05)(0.99)} = \frac{0.009}{0.009+0.0495} \approx 0.155$$



BAYES' THEOREM EXAMPLE: DEFECT DETECTION

- **Interpretation**
 - Even though the test is “90% accurate,” a flagged item is only about 15.5% likely to be truly defective, primarily because defects are very rare (only 1%).
 - Key Insight: When the “prior” probability $P(A)$ is small (defects are rare), false positives can overwhelm true positives.
 - **Takeaway:**
 - Always consider **base rates** or **prior probabilities** when interpreting test results.
 - This example directly illustrates why Bayes' Theorem is crucial in quality control, medical tests, and more.
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DISCRETE RANDOM VARIABLES

- **Reference:** Walpole et al. (9th ed.), Chapters 3.1–3.2
 - What Is a Random Variable?
 - A variable whose value depends on the outcome of a chance experiment.
 - Discrete means it takes on a countable set of values (finite or infinite).
 - Examples: X = number of defective products, X = number of heads in 10 coin tosses.
 - Probability Mass Function (pmf)
 - Denoted $p(x)=P(X=x)$.
 - Must satisfy
 1. $p(x)\geq 0$ for all x .
 2. $\sum p(x)=1$ (sum of probabilities over all possible x equals 1).
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DISCRETE RANDOM VARIABLES

- Cumulative Distribution Function (CDF)
 - $F(x)=P(X\leq x)$.
 - Summation of pmf values from the smallest possible outcome up to x .
 - Real-World Illustration
 - Quality Control: Let X = number of defective items in a sample of size n .
 - Reliability: Let X = number of breakdowns in a machine during a month.
 - Biology: Let X = number of individuals with a certain genetic trait in a group.
 - Why This Matters
 - Discrete random variables help you calculate probabilities of counts (e.g., “Exactly 2 defective items” or “At most 1 breakdown”).
 - Fundamental to many discrete probability distributions: Bernoulli, Binomial, Geometric, Hypergeometric, and Poisson.
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MEAN, VARIANCE, AND STANDARD DEVIATION

- **Reference:** Walpole et al. (9th ed.), Chapters 4.1–4.4
 - **Expected Value (Mean, μ)**
 - $E[X] = \sum x p(x)$, for discrete X .
 - Interpreted as the “long-run average” of X if the experiment is repeated many times.
 - **Variance (σ^2)**
 - $Var(X) = E[X^2] - (E[X])^2$
 - Captures the “spread” or “dispersion” of X around its mean. A larger variance indicates values of X fluctuate more widely.
 - **Standard Deviation (σ)**
 - $\sigma = \sqrt{Var(X)}$
 - Uses the same units as X , so it’s more intuitive than variance.
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MEAN, VARIANCE, AND STANDARD DEVIATION

- **Examples**

- Coin Toss (Binomial): If X = # of heads in 10 fair tosses, then
 - $E[X]=10 \times 0.5=5$ ($n p$), $\text{Var}(X)=10 \times 0.5 \times 0.5=2.5$ ($n p (1-p)$). $\sigma \approx 1.58$
- Hospital Arrivals (Poisson): If X = # of arrivals per hour with rate $\lambda=4$, then
 - $E[X]=4$, $\text{Var}(X)=4$ (property of the Poisson: both mean and variance are equal to λ), $\sigma=2$.

- **Key Properties**

- Linearity: $E[aX+b] = a E[X]+b$.
- Scaling of Variance: $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

- **Intuitive Interpretation**

- Mean: “Center” or average outcome.
 - Standard Deviation: Typical distance from the center.
 - Important for decision-making, quality control, risk analysis, etc.
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DISCRETE DISTRIBUTIONS

- **Reference:** Walpole et al. (9th ed.), Chapters 5.1–5.5
 - **Binomial Distribution**
 - Scenario
 - We have n independent Bernoulli trials (e.g., flipping a coin or checking if a product is defective).
 - Each trial has only two outcomes (“success” with probability p or “failure” with probability $1-p$).
 - Random Variable
 - X = number of successes in n trials.
 - Probability Mass Function (pmf)
 - $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, k = 0, 1, 2, \dots, n.$
 - $\binom{n}{k}$ is the binomial coefficient, “the number of ways to choose “ k ” successes out of “ n ”.”
 - Mean & Variance
 - $E[X] = n p, \text{ Var}(X) = n p(1-p).$
 - Real-World Use
 - Quality Control: # of defectives in a batch of size n .
 - Clinical Trials: # of patients responding to a new drug out of n .
 - Key Assumption: Trials are independent, and the probability of success p stays the same in every trial.
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DISCRETE DISTRIBUTIONS

- **Geometric Distribution**

- Scenario
 - We have repeated independent Bernoulli trials, each with success probability p .
 - We keep going **until** we see the first success.
 - Random Variable
 - Y = the trial count on which the first success occurs.
 - Thus $Y \in \{1, 2, 3, \dots\}$.
 - Probability Mass Function (pmf)
 - $P(Y = k) = p (1 - p)^{k-1}$, $k = 1, 2, \dots$
 - Mean & Variance
 - $E[Y] = \frac{1}{p}$ $Var(Y) = \frac{(1-p)}{p^2}$.
 - Real-World Use
 - Coin Tossing: # of flips until the first head appears.
 - Machine Lifetime: # of cycles until first breakdown (if each cycle independently has the same chance of breakdown).
 - Key Note: Geometric is a special case of the Negative Binomial distribution (when we only need 1 success).
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DISCRETE DISTRIBUTIONS

- **Negative Binomial Distribution**
 - Scenario
 - Generalizes the Geometric idea. We run Bernoulli trials until we get r successes (for some fixed r , e.g., 2 or 3).
 - Probability of success in each trial is p .
 - Random Variable
 - W = total number of trials needed to achieve r successes.
 - Probability Mass Function (pmf)
 - $P(W = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, k = r, r+1, r+2, \dots$
 - Mean & Variance
 - $E[W] = \frac{r}{p} \quad \text{Var}(W) = \frac{r(1-p)}{p^2}.$
 - Example
 - **Product Testing:** “How many parts do we test until we find r defective ones?”
 - The Negative Binomial includes the Geometric when $r=1$.
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DISCRETE DISTRIBUTIONS

- **Hypergeometric Distribution**

- Scenario
 - We have a finite population of size N , containing D “success” items (e.g., defectives).
 - We draw n items without replacement.
 - Random Variable
 - X = # of success items found in our n -draw sample.
 - Probability Mass Function (pmf)
 - $$P(X = k) = \frac{\binom{D}{k} \binom{N-D}{n-k}}{\binom{N}{n}}, \quad k = 0, 1, 2, \dots, n$$
 - Mean
 - $E[X] = n \frac{D}{N}$.
 - Example
 - Acceptance Sampling: Checking # of defective items in a random sample from a batch (no replacement!).
 - Key Contrast: Differs from the Binomial distribution because p is not constant (drawing without replacement changes the fraction of “successes” left in the population).
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DISCRETE DISTRIBUTIONS

- **Poisson Distribution**

- Scenario

- Models the number of events occurring in a given time or space interval.
- Good approximation to Binomial when n is large and p is small, so $\lambda = np$ is moderate.

- Random Variable

- $X = \#$ of occurrences/events in a specified interval.

- Probability Mass Function (pmf)

- $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$

- Mean & Variance

- $E[X] = \lambda, \quad Var(X) = \lambda$.

- Example

- **Queueing**: # of arrivals per hour at a call center.
- **Defect Counts**: # of flaws per yard of fabric.

- Observation: Poisson distribution's equal mean and variance (λ) is a key hallmark.

DISCRETE DISTRIBUTIONS SUMMARY

- **Which Distribution to Use?**

1. **Binomial**: n independent trials, same p , count # successes.
 2. **Geometric/Negative Binomial**: # of trials until 1 or r successes.
 3. **Hypergeometric**: Like Binomial but without replacement (finite population).
 4. **Poisson**: Count of events over time/space; also an approximation to Binomial for large n but small p .
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USING CUMULATIVE DISTRIBUTION TABLES

- **Why Tables?**

- For Binomial, Poisson, others—tables help find $P(X \leq k)$ quickly.

- **Software & Tools**

- Statistical tables in Walpole (Appendix).
- R/Python: e.g., `pbinom`, `ppois` functions.
- Online calculators (use carefully and check correctness).

- **Example**

- Suppose $X \sim \text{Poisson}(\lambda=4)$. We want $P(X \leq 3)$.
 - Use Poisson table or software to sum up $P(X=0)+P(X=1)+P(X=2)+P(X=3)$.
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EXERCISE 1

A factory produces electronic components with a **5% defect rate**. Suppose we randomly select **$n=10$** components. Let X be the number of defectives among these 10.

- 1.(a) Find $P(X=2)$, the probability exactly 2 out of 10 are defective.
- 2.(b) Find $P(X \leq 3)$, the probability at most 3 out of 10 are defective.

Distribution?

Binomial Distribution

EXERCISE 1

- Solution:
 - Distribution and Parameters
 - $X \sim \text{Binomial}(n=10, p=0.05)$.
 - Pmf: $P(X = k) = \binom{10}{k} (0.05)^k (1 - 0.05)^{10-k}$
 - (a) Calculating $P(X = 2)$
 - $P(X = 2) = \binom{10}{2} (0.05)^2 (0.95)^{10-2}$
 - $\binom{10}{2} = \frac{10!}{2! 8!} = 45$
 - $P(X = 2) = 45 * (0.05)^2 * (0.95)^8 = 0.0746$
 - (b) Calculating $P(X \leq 3)$
 - $P(X \leq 3) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$.
 - Compute each term similarly:
 - $P(X = 0) = \binom{10}{0} (0.05)^0 (0.95)^{10-0} = 0.5987$
 - $P(X = 1) = \binom{10}{1} (0.05)^1 (0.95)^{10-1} = 0.3151$
 - $P(X = 2) = \binom{10}{2} (0.05)^2 (0.95)^{10-2} = 0.0746$
 - $P(X = 3) = \binom{10}{3} (0.05)^3 (0.95)^{10-3} = 0.01$

Sum them up:

$$P(X \leq 3) \approx 0.5987 + 0.3151 + 0.0746 + 0.0100 = 0.9984.$$

So there's about a 99.84% probability that at most 3 components are defective out of 10.

EXERCISE 2

- A basketball player has a **free-throw success probability** of $p=0.8$. Let Y be the trial (shot) count on which she **makes her first basket**. We assume each shot is independent.
- 1.(a) What is $P(Y=3)$? (i.e., the probability that she succeeds for the first time on her 3rd shot)
 - 2.(b) What is the expected shot number $E[Y]$ at which she makes her first basket?

Distribution?

Geometric

EXERCISE 2

- **Identify Distribution**

- $Y \sim \text{Geometric}(p=0.8)$.
- pmf: $P(Y = k) = p (1 - p)^{k-1}$.

- **(a) Probability**

- $P(Y = 3) = 0.8 (1 - 0.8)^{3-1} = 0.8 * 0.2^2 = 0.032$

- **(b) Expectation**

- $E[Y] = \frac{1}{p} = \frac{1}{0.8} = 1.25$

- So, on **average**, she'll succeed on her **1.25th** shot. Interpreted practically, this means she usually scores on her **first** shot, but occasionally it takes a second or third.
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EXERCISE 3

- We continue with the same basketball player ($p=0.8$ per shot). Let W be the total number of shots she needs to make **3 successful free throws**.
- 1.(a) What is $P(W=5)$? (The probability that her 3rd success occurs exactly on the 5th shot)
 - 2.(b) Find $E[W]$, the average number of total shots required to get 3 successes.

Distribution?

Negative Binomial Distribution

EXERCISE 3

- **Distribution**

- $W \sim \text{Negative Binomial}(r=3, p=0.8)$.
- pmf :

- $P(W = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r+1, r+2, \dots$

- **(a) Probability** ($k=5, r=3$)

- $P(W = 5) = \binom{5-1}{3-1} (0.8)^3 (1-0.8)^{5-3}$
- $\binom{4}{2} = 6$
- $P(W = 5) = 6 * (0.8)^3 * (0.2)^2 = 0.12288$

- **(b) Mean**

- $E[W] = \frac{r}{p} = \frac{3}{0.8} = 3.75$

- So on average, she needs about 3.75 shots to get 3 successful free throws.
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EXERCISE 4

- We have a lot of **40** semiconductor chips, of which **5** are defective. We choose **6** chips **without replacement**. Let X = number of defective chips in the chosen 6.
- 1.(a) What is $P(X=2)$? (The probability that exactly 2 of the 6 chosen are defective)
 - 2.(b) What is $E[X]$, the expected number of defective chips in our 6?

Distribution?

Hypergeometric

EXERCISE 4

- **Distribution**

- $X \sim \text{Hypergeometric}(N=40, D=5, n=6)$.
- Pmf:

- $$P(X = k) = \frac{\binom{5}{k} \binom{40-5}{6-k}}{\binom{40}{6}}$$

- **(a) Compute $P(X=2)$**

- $$P(X = 2) = \frac{\binom{5}{2} \binom{35}{4}}{\binom{40}{6}} = \frac{10 \cdot 52360}{3838380} = 0.1364$$

- **(b) Mean**

- $$E[X] = n \frac{D}{N} = 6 * \frac{5}{40} = 0.75$$

- On average, you get 0.75 defective chips (i.e., about 3 out of 4 times on average you pick 1 defective if 5 are in the entire lot).
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EXERCISE 5

- Defects in a woven textile occur at an **average rate** of $\lambda=2$ per 100 meters. Assume defects follow a Poisson process. Let X = number of defects in a **100-meter roll**.
 - 1.(a) What is $P(X=3)$? (The probability of exactly 3 defects in 100 meters)
 - 2.(b) What is $P(X \geq 2)$? (The probability of 2 or more defects)
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EXERCISE 5

- **Distribution**

- $X \sim \text{Poisson}(\lambda=2)$.
- Pmf:

- $$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

- **(a) Calculating $P(X=3)$**

- $$P(X = 3) = \frac{e^{-2} 2^3}{3!} = \frac{4 \cdot 0.1353}{3} = 0.1804$$

- **(b) Calculating $P(X \geq 2)$**

- $P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$.

- $$P(X = 0) = \frac{e^{-2} 2^0}{0!} = 0.1353$$

- $$P(X = 1) = \frac{e^{-2} 2^1}{1!} = 0.2707$$

- $P(X \geq 2) = 1 - (0.1353 + 0.2707) = 1 - 0.4060 = 0.5940$

- So there is about a 59.4% chance that we will see at least 2 defects in 100 meters.
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